algorithm are very moderate and the exponential worst-case memory requirements do not seem to manifest themselves in practice. Further evidence for this provides the fact that, so far at least, we have not received any issue from a user reporting extensive memory consumption when running Concuerror.

12 RELATED WORK

In early approaches to stateless model checking, it was observed that reduction was needed to combat the explosion in number of explored interleavings. Consequently, since then, several reduction methods have been proposed including partial order reduction and bounding techniques such as bounding the depth [Godefroid 1997], the number of context switches and the number of preemptive context switches [Musuvathi and Qadeer 2007], or the number of delays that an otherwise deterministic scheduler is allowed [Emmi et al. 2011]. Since early persistent set techniques [Clarke et al. 1999; Godefroid 1996; Valmari 1991] relied on static analysis, sleep set techniques were also used to dynamically prevent explorations from processes that would be provably redundant. However, it was observed that, although sleep sets are sufficient to prevent the complete exploration of different but equivalent interleavings [Godefroid et al. 1995], additional techniques were needed to reduce sleep set blocked exploration.

The dynamic partial order reduction algorithm of Flanagan and Godefroid [2005b] showed how to construct persistent sets on-the-fly "by need", leading to better reduction. Similar techniques have been combined with dynamic symbolic execution, which is also known as *concolic testing*, where new test runs are initiated in response to detected races by flipping these races using postponed sets [Sen and Agha 2007]. Since then, several variants, improvements, and adaptations of DPOR for stateless model checking [Lauterburg et al. 2010; Tasharofi et al. 2012] and concolic testing [Saarikivi et al. 2012; Sen and Agha 2006] have appeared, all based on persistent sets. Our source-DPOR and optimal-DPOR algorithms can be applied to all these contexts to provide increased or optimal reduction in the number of explored interleavings.

A related area is *reachability testing*, in which test executions of concurrent programs are steered by the test harness. Lei and Carver [2006] present a technique for exploring all Mazurkiewicz traces in a setting with a restricted set of primitives (message passing using FIFO channels and monitors) for process interaction. The scheduling of new test executions explicitly pairs message transmissions with receptions, and could potentially require significant memory, compared to the more light-weight approach of stateless model checking. The technique of Lei and Carver guarantees to avoid re-exploration of different but equivalent maximal executions (corresponding to Theorem 7.11), but reports blocked executions. Moreover, their technique requires a non-trivial amount of memory for storing interleavings that are yet to be explored.

Kahlon et al. [2009] present a normal form for executions of concurrent programs and prove that two different normal-form executions are not in the same Mazurkiewicz trace. This normal form can be exploited by SAT- or SMT-based bounded model checkers, but it can *not* be used by stateless model checkers that enumerate the execution sequences by state-space exploration. Kähkönen et al. [2012], and very recently Rodríguez et al. [2015], use unfoldings [McMillan and Probst 1995], which can also obtain optimal reduction in number of interleavings. However, unfolding-based techniques have significantly larger cost per test execution than DPOR-like techniques, and the technique of Kähkönen et al. [2012] also needs an additional post-processing step for checking non-local properties such as races and deadlocks. A technique for using transition-based partial order reduction for message-passing programs, without moving to an event-based formulation is to refine the concept of dependency between transitions to that of *conditional dependency* [Godefroid 1996; Godefroid and Pirottin 1993; Katz and Peled 1992].